COMPUTER VISION / INDUSTRIAL QUALITY

LineSight

Visual anomaly detection for production lines

A complete CPU-first inspection project built around a normal-only Isolation Forest, real industrial bottle images, a local inspection application, and auditable experiment evidence.

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HELD-OUT AUROC	DEFECT RECALL	PRECISION
0.9468	84.13%	96.36%

The real-world problem

Bottle production needs to identify breakage and contamination even when labeled defect examples are scarce. A fixed camera can collect normal parts; LineSight learns their visual pattern and flags unusual images for an operator to review. This benchmark models that inspection task; it does not represent an installed factory system.

What is delivered

Verified dataset preparation; four development candidates; frozen calibrated model; held-out predictions and uncertainty; five result figures; upload/replay/camera dashboard; FastAPI and SQLite; tests, dependency audit, CI, container recipe, executed notebook, model/dataset cards, and GitHub-ready local source.

How to read the results

The selected model detected **53 of 63 defects**, missed **10**, and flagged **2 of 20 normal bottles**. It requires a controlled view and lighting. Camera/lighting changes produced substantial false alarms in sensitivity probes. No claim of perfection, general-purpose vision, or hard real-time factory performance is supported.

Data, architecture and model logic

Data: MVTec AD bottle, 292 original 900 x 900 images and 63 defect masks. Acquisition uses a pinned public mirror. Its archive digest, structural counts, per-image digests and duplicate checks were verified; official-download byte equivalence was not independently established. The original official train/test roles are retained.

Partition	Images	Role
Fit	125 normal	Learn median/MAD reference and fit forest
Tune	42 normal + 126 proxy variants	Compare candidates using synthetic defects only
Calibration	42 normal	Set independent score threshold
Official test	20 normal + 63 defective	Evaluate once after design/threshold freeze

Offline and online architecture

Stage	Flow
Offline	Pinned images > audited manifest > fit-only reference > candidate forests > tune proxy selection > independent calibration > frozen artifact > held-out evaluation
Online	Upload / recorded replay / sampled camera > bounded image validation > fixed descriptors > Isolation Forest > score / threshold / coarse evidence > operator view and SQLite event

Feature construction and detector

Inputs are normalized to RGB and resized to 192 x 192. Robust per-pixel median and MAD statistics are learned from fit normals. Color and gradient residuals, brightness and edge statistics form 16 descriptors over each of 144 non-overlapping 16 x 16 patches. The selected forest contains 192 trees with 512 samples per tree. Each image receives the maximum negative `score_samples` patch score.

Isolation trees randomly split features. Unusual feature vectors tend to isolate with shorter paths. Larger exposed scores mean more anomalous; they are not probabilities. The 12 x 12 evidence grid is a coarse score map, not a validated segmentation output.

Development, calibration and reproducibility

Candidate development used tune normals and deterministic scratches, blobs and inverted patches derived only from those tune images. No real test defect or test score selected the model or threshold. Proxy performance is a development signal, not real-defect evidence.

Candidate	Proxy AUROC	Selected
image_if	0.8152	
patch_if_max	0.9211	Yes
patch_if_top4	0.8643	
patch_if_top9	0.8054	

Independent nominal threshold

For n = 42 calibration normals and alpha = 0.05, the upper rank is ceil((n + 1) x (1 - alpha)) = 41. The frozen threshold is **0.73856191**, with a strict score > threshold decision. The finite-sample nominal error bound is 4.65% under exchangeability. This is a marginal nominal coverage statement, not a promise about this small test set or a shifted camera domain.

Audit trail

Model identifier: **ls-a5006862e1ce**. The frozen artifact and dataset manifest digests are checked before evaluation. Selection history, proxy descriptions, seed rules, normal calibration scores and all 83 test predictions are stored in CSV/JSON files. The random seed is 42; uncertainty resampling uses seed 2026.

Original fit environment: Python 3.12.14, scikit-learn 1.8.0, Pillow 12.3.0. Deployment runtime pins Pillow 13.0.0 after a dependency-security correction. The frozen artifact is retained; a separate verification checks all original test scores against the updated image library without changing selection.

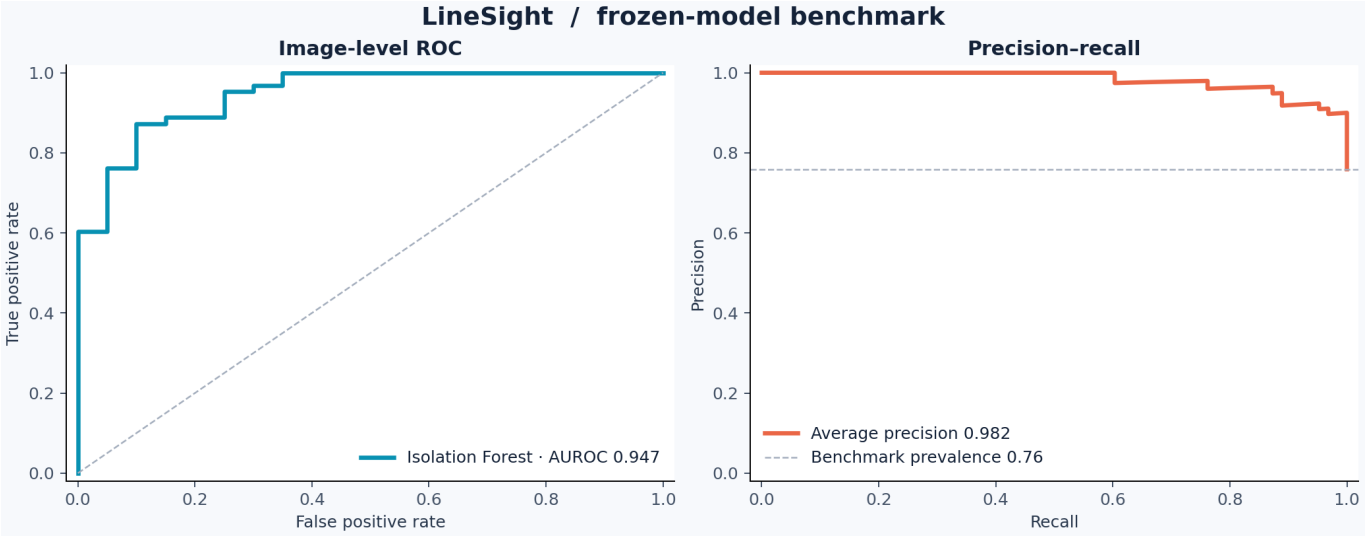
Runtime verification: Pillow 13.0.0: all 83 frozen test decisions match; maximum score difference 0.

Licensing

Original code is MIT licensed. The MVTec dataset and supplied dataset-derived visuals and pretrained material are provided under CC BY-NC-SA 4.0 for noncommercial work. Commercial deployment needs separately licensed data and retraining.

Held-out performance and uncertainty

Metric	Observed	95% stratified bootstrap interval
AUROC	0.9468	0.8857 - 0.9865
Average precision	0.9819	0.9611 - 0.9958
Precision	0.9636	0.9123 - 1.0000
Defect recall	0.8413	0.7460 - 0.9206
F1	0.8983	0.8364 - 0.9500

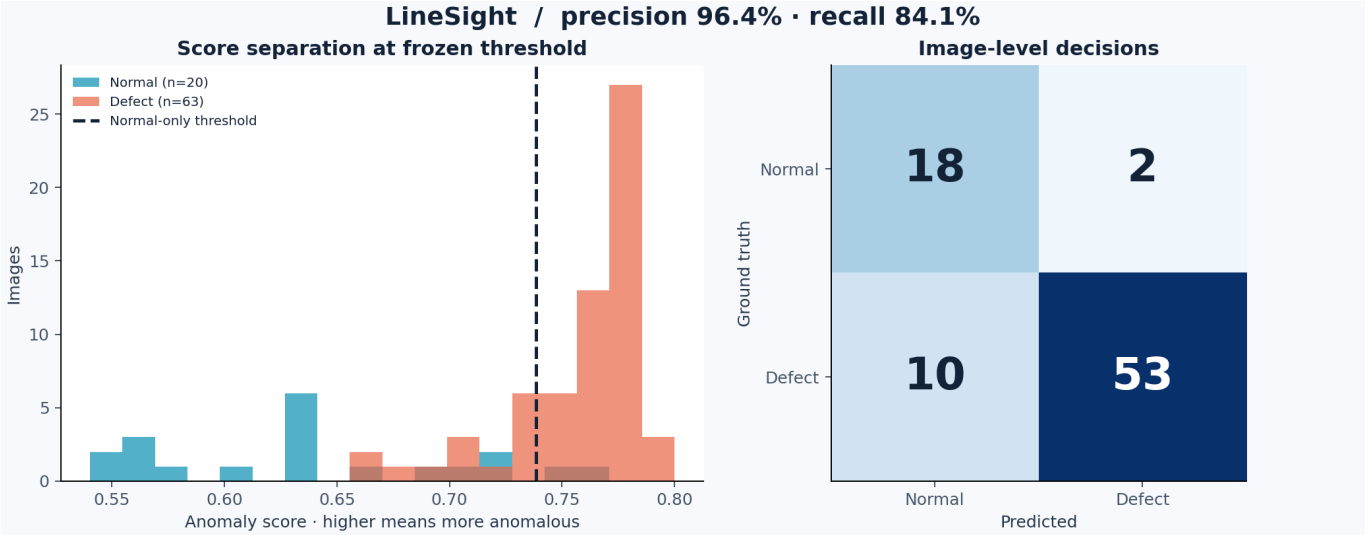


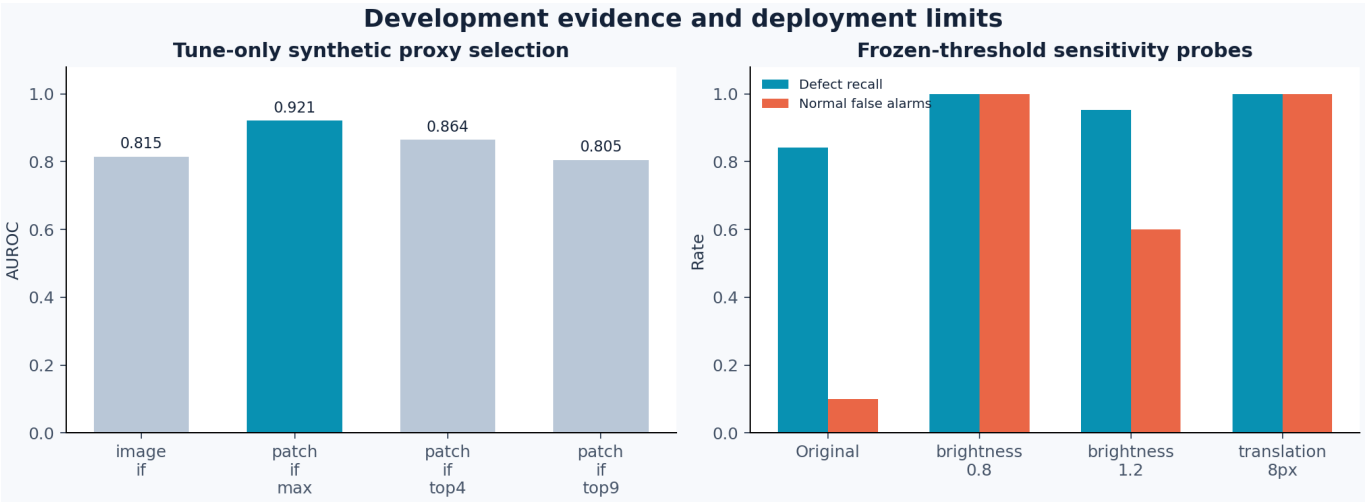
Intervals use 2,000 class-stratified percentile bootstrap samples. They condition on this small, defect-heavy benchmark mix; they do not estimate factory generalization or production precision at a low defect prevalence.

Real defect	Detected / total	Recall
broken large	20 / 20	100.00%
broken small	19 / 22	86.36%
contamination	14 / 21	66.67%

A simple fit-reference residual baseline achieved AUROC 0.9389, recall 76.19%, and F1 0.8649. Isolation Forest provides a modest observed improvement on this split; no significance or state-of-the-art claim is made.

Decisions, misses and sensitivity



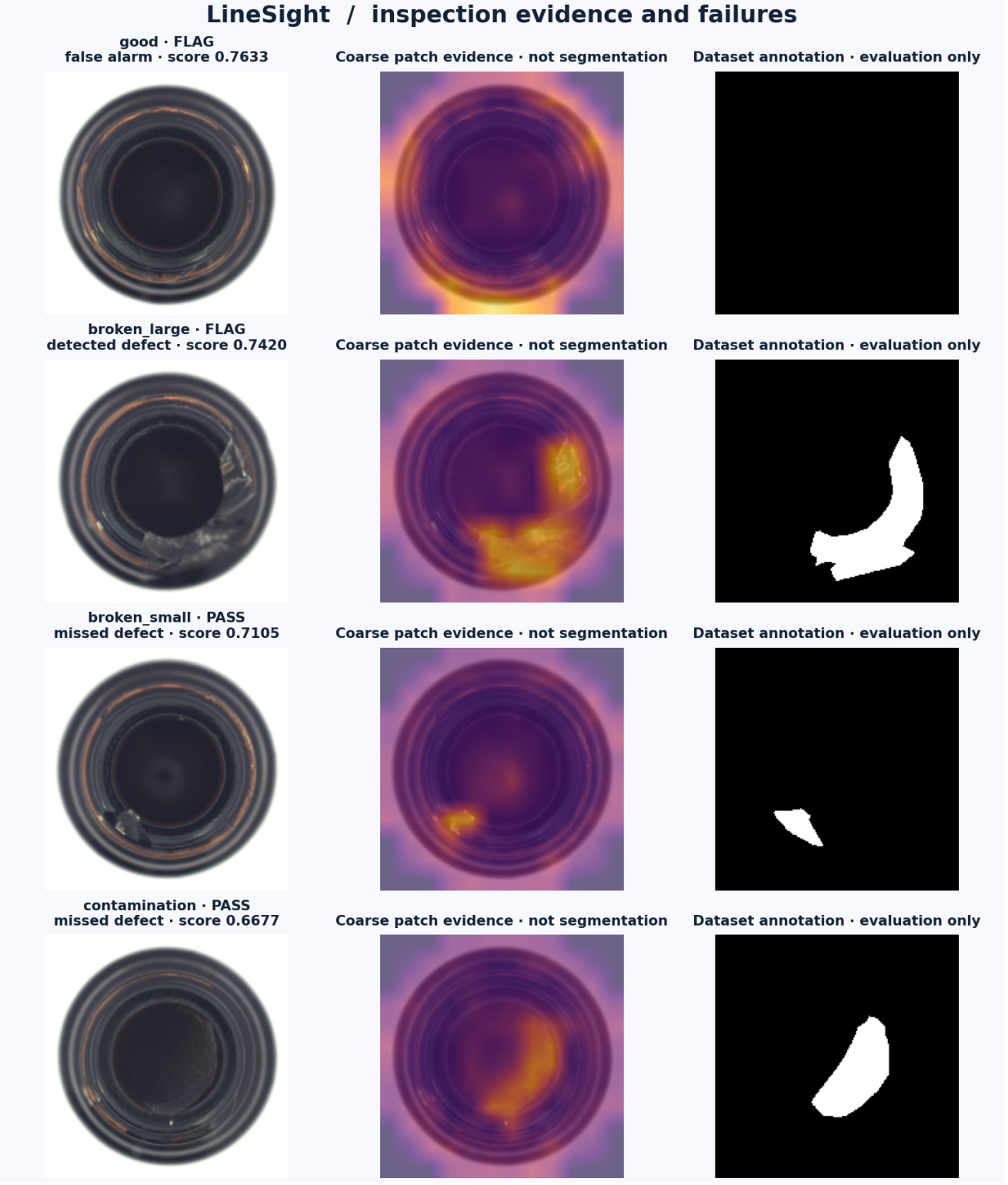


The fixed threshold missed 10 defective images, especially contamination (14 of 21 detected), and produced two false alarms. Reducing brightness to 80% or translating the resized view by eight pixels flagged every normal test image. These are controlled post-freeze sensitivity probes, not additional model-selection runs. Correct camera alignment and stable illumination are operational prerequisites.

Condition	Defect recall	Normal false alarms
brightness 0.8	100.00%	100.00%
brightness 1.2	95.24%	60.00%
translation 8px	100.00%	100.00%

Visual inspection evidence

Reproducible representatives are shown with their decision score, coarse patch evidence and the dataset annotation. When a decision error exists it is promoted into the example selection. The annotation is not consumed by training or the deployed detector.



Dataset-derived images: MVTec AD, CC BY-NC-SA 4.0. No pixel segmentation metric is claimed.

Application, tests and deployment path

The local dashboard sends actual images to the same frozen detector used in evaluation. It supports uploads, clearly labeled recorded replay, sampled browser-camera frames, a coarse evidence overlay and a bounded local decision history. SQLite retains at most 1,000 metadata records. Uploaded images are not retained.

Verification area	Evidence / behavior
Model and data	Partition isolation, all source digests, calibration rank and nominal error math, score orientation, deterministic persistence, stored prediction reconciliation
API and security	Missing-model readiness, image bytes/pixels/formats, bounded inspection requests, invalid inputs, history retention, sample path confinement, API reference, real detector integration
Application	Managed browser review of dashboard, dataset inspection and history; physical camera validation unavailable
Dependencies	Pinned runtime and recorded advisory audit; image library upgraded with frozen-score equivalence check
Deployment	Native local service verified. Nonroot Docker/Compose recipe supplied; container build was not run because Docker is unavailable
Repository	Local main branch, owner-only Git identity, CI recipe, no remote, no push or external publication

Recorded validation: 50 tests passed; zero failures, errors or skips. Eight notebook cells executed. Browser replay, decisions, history and API reference verified. Runtime audit: zero known advisories in 30 pinned packages.

Measured runtime

Original benchmark CPU inference p50 was 139.21 ms and p95 324.66 ms, batch size one, one forest thread, across 83 images after two warmups. It includes resizing, features and forest scoring; excludes image decode, HTTP, capture, queueing and SQLite. Shared-host load affects these timings. Browser camera sampling targets about 1.2 seconds between completed requests. This is a soft inspection demonstration, not a validated line-speed or hard real-time guarantee.

Reproduction, handoff and next validation

Task	Command / location
Install	python -m pip install -r requirements.lock python -m pip install -e '.[dev]'
Acquire / validate data	python scripts/download_data.py python scripts/download_data.py --verify-only
Train / freeze	python scripts/train.py
Evaluate	python scripts/evaluate.py
Tests	python -m pytest -q
Use dashboard	python scripts/serve.py http://127.0.0.1:8000
Notebook	notebooks/01_linesight_walkthrough.ipynb
Read model / data details	docs/model-card.md; docs/dataset-card.md
Inspect exact evidence	reports/metrics.json; test_predictions.csv; frozen_model.json; runtime_verification.json; pytest.xml
Build delivery archives	python scripts/package_release.py

Before a real production pilot

Collect licensed normal images from the actual camera, product, alignment and lighting. Separate batches or acquisition periods when splitting data. Gather independent defect examples and enough normal images to measure operational false alarms. Define acceptable miss/false-alarm costs, review queues, drift monitoring and operator escalation. Add authentication, TLS, resource/time limits and observability before network exposure. Physical rejector integration requires separate hardware timing and safety validation.

Sources and attribution

Bergmann, P., Fauser, M., Sattlegger, D., and Steger, C. (2019). MVTec AD - A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection. CVPR. Official dataset: www.mvtec.com/research-teaching/datasets/mvtec-ad.

Isolation Forest implementation: scikit-learn, scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html. Pinned acquisition, licenses and primary reference links are recorded in the repository.

Project owner: Kumar Waibhav Akshat | kwa.waibhav@gmail.com. The supplied repository is prepared locally for review and future GitHub publication; nothing has been pushed or published externally.